

Data Science x Cancer Pharmacogenomics

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Spring 2026

Data Science Capstone



Introduction

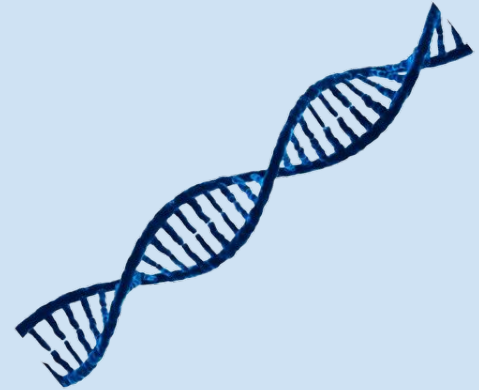


The Problem

Cancer drug response is not often combined with personalized plans for patients who have specific cancer types and genomic profiles, leading to long-term treatment plans with less efficacy.



Can we predict how sensitive a cancer cell line
will be to a drug given its biological and
pharmacological profile?



Objectives

Predict the sensitivity of cancer cell lines to cancer drugs using cell lines biological and pharmacological profile.

Assist healthcare professionals in prescribing to patients for more effective treatment plans

Use various modeling techniques to find the most effective and accurate method for reducing the size of cancerous tumors

Data Acquisition



Data Acquisition

What was the source of our data?

- The *Genomics of Drug Sensitivity in Cancer, or GDSC, project*
 - A collaboration by the Sanger Institute (UK) and the Massachusetts General Hospital Cancer Center (USA)
 - Took downloads directly from the GDSC database
 - Utilized a merged *Kaggle* dataset
- The *Cancer Cell Line Encyclopedia, or CCLE, project*
 - A collaboration between the Broad Institute and the Novartis Institutes for Biomedical Research
 - Was used to cross-fill missing or weak values

Data Acquisition

How was data collected?

- The GDSC project involved a large-scale screening of many different human cancer cell lines treated with a wide range of anti-cancer drugs.
 - Cancer cells were seeded in plates under optimal growth conditions
 - 24 hours after plating, cells were treated with a dose titration of each drug
 - After drugging, plates were incubated and cell viability was measured after 72 hours
- Information about the cancer cell line, the drug, was recorded as well as the IC₅₀, which measures the effectiveness of the drug in reducing cell viability by indicating the sensitivity of each cancer cell line to the drug administered.

Data Acquisition

Feature	Description	Role	Source
LN_IC50	Natural log of the half-maximal inhibitory concentration (IC50), used as the main drug-response target.	TARGET	GDSC (K)
DRUG_NAME	Name of the drug tested on the cell line.	Identifier	GDSC (K)
TCGA_DESC	Cancer type label based on The Cancer Genome Atlas abbreviation system.	Feature (IMP)	GDSC (K) with CCLE
GDSC Tissue descriptor 1	Broad tissue or organ-system category for the cell line.	Feature (IMP)	GDSC (K)
GDSC Tissue descriptor 2	More specific tissue or disease subtype within the broader tissue category.	Feature (IMP)	GDSC (K)
Growth Properties	Describes how the cell line grows in culture, such as adherent or suspension.	Feature	GDSC (K)
TARGET_PATHWAY	The biological pathway associated with the drug's target or mechanism of action.	Feature	GDSC (K)
Microsatellite instability Status (MSI)	Indicates whether the cell line shows microsatellite instability status, such as MSI or stable forms. The source paper explicitly includes MSI as one of the genomic factors analyzed in relation to drug response.	Feature	GDSC (K)
mutational_burden	a biological feature describing how many mutations are present in the model/cell line.	Feature	GDSC
ploidy_snp6	a ploidy estimate derived from SNP6 array data.	Feature	GDSC
ploidy_wes	a ploidy estimate derived from whole-exome sequencing data.	Feature	GDSC

❑ Target

❑ Pharmacological

❑ Biological

❑ Genomic

Data Acquisition

How this data was used to attain the objectives of our project?

- Our primary objective: to develop a model that can predict the sensitivity of cancer cell lines to different drugs using genotypic, phenotypic, and pharmacological information
- This data provides the necessary information to build a predictive model based on those factors that predicts the effectiveness of a cancer drug
- This model then allows us to build a tool that can use such information to predict optimal treatment specific to a patient

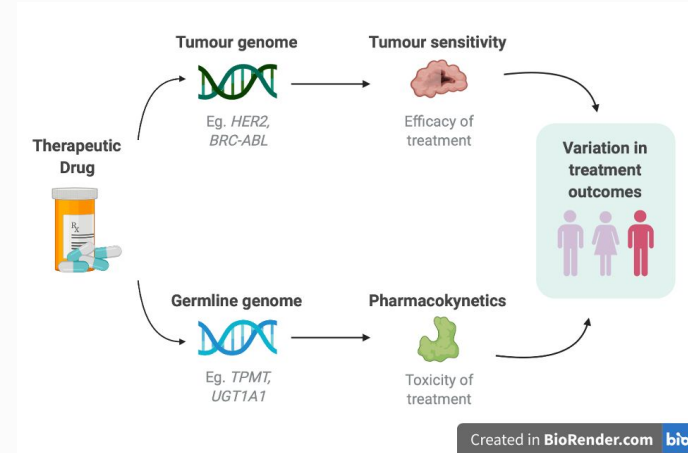
Data Science Aspect



Data Science Aspect

- ❖ Improve quality of life for people with cancer.
- ❖ Enable precision medicine
- ❖ LN_IC50 values used to measure the sensitivities.
- ❖ Enhancing clinical trial design
- ❖ Manage adverse effects
- ❖ Matching the right drugs to the right patient

What is the relevance of doing this project?

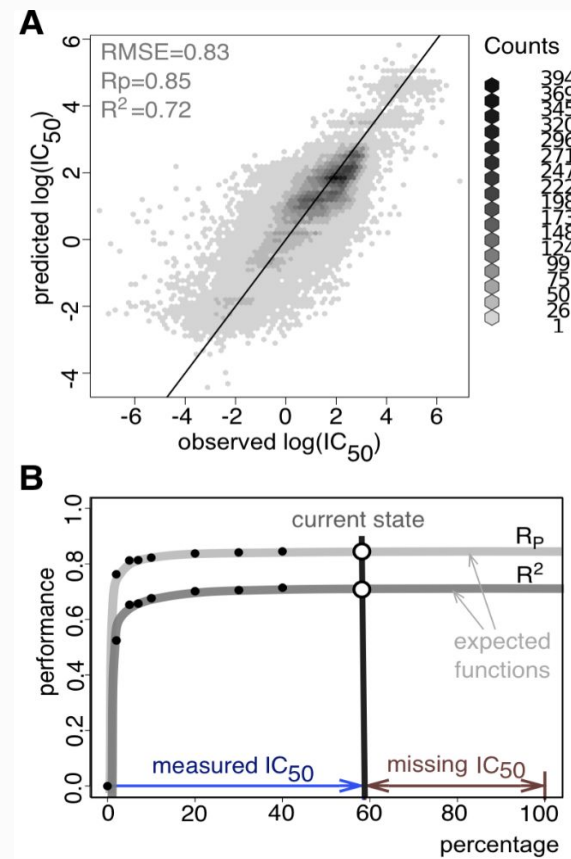


> <https://link.springer.com/article/10.1186/s12943-025-02400-z>

<https://journals.plos.org/plosone/article?id=10.1371/journal.pone.0061318>

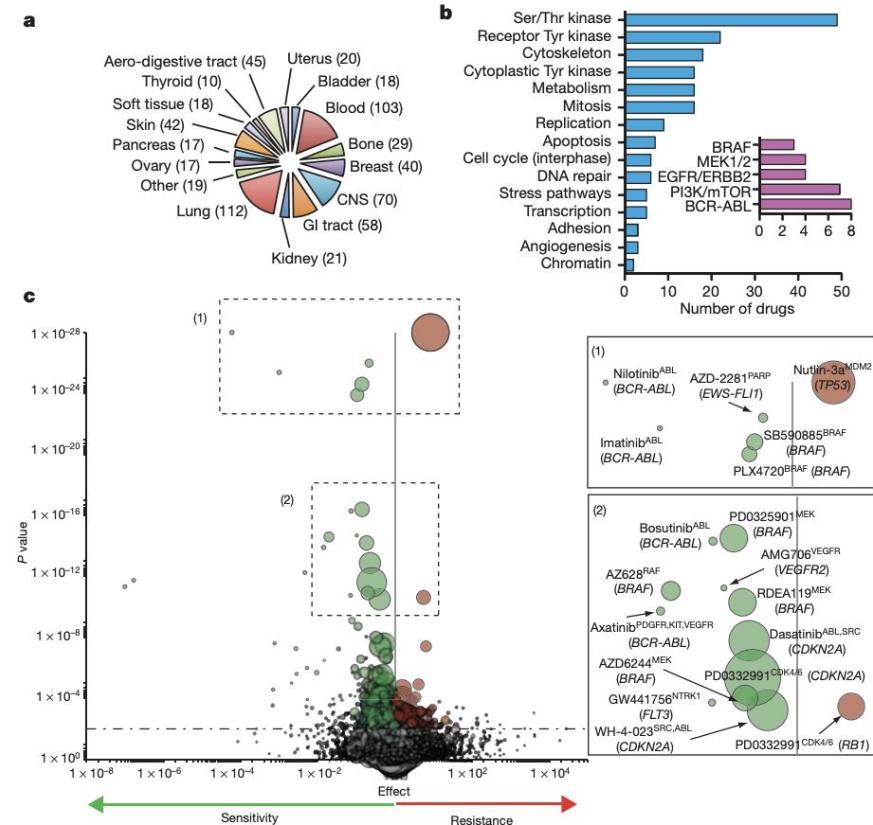
Relevant work

- ❖ “Machine Learning Prediction of Cancer Cell Sensitivity to Drugs Based on Genomic and Chemical Properties” Michael P. Menden et al.
 - Machine learning can be used to predict cell sensitivity to certain treatment types.



Relevant Work Continued...

- ❖ “Systematic identification of genomic markers of drug sensitivity in cancer cells” Matthew Garnett et al.
- Biomarker discovery and Patient stratification



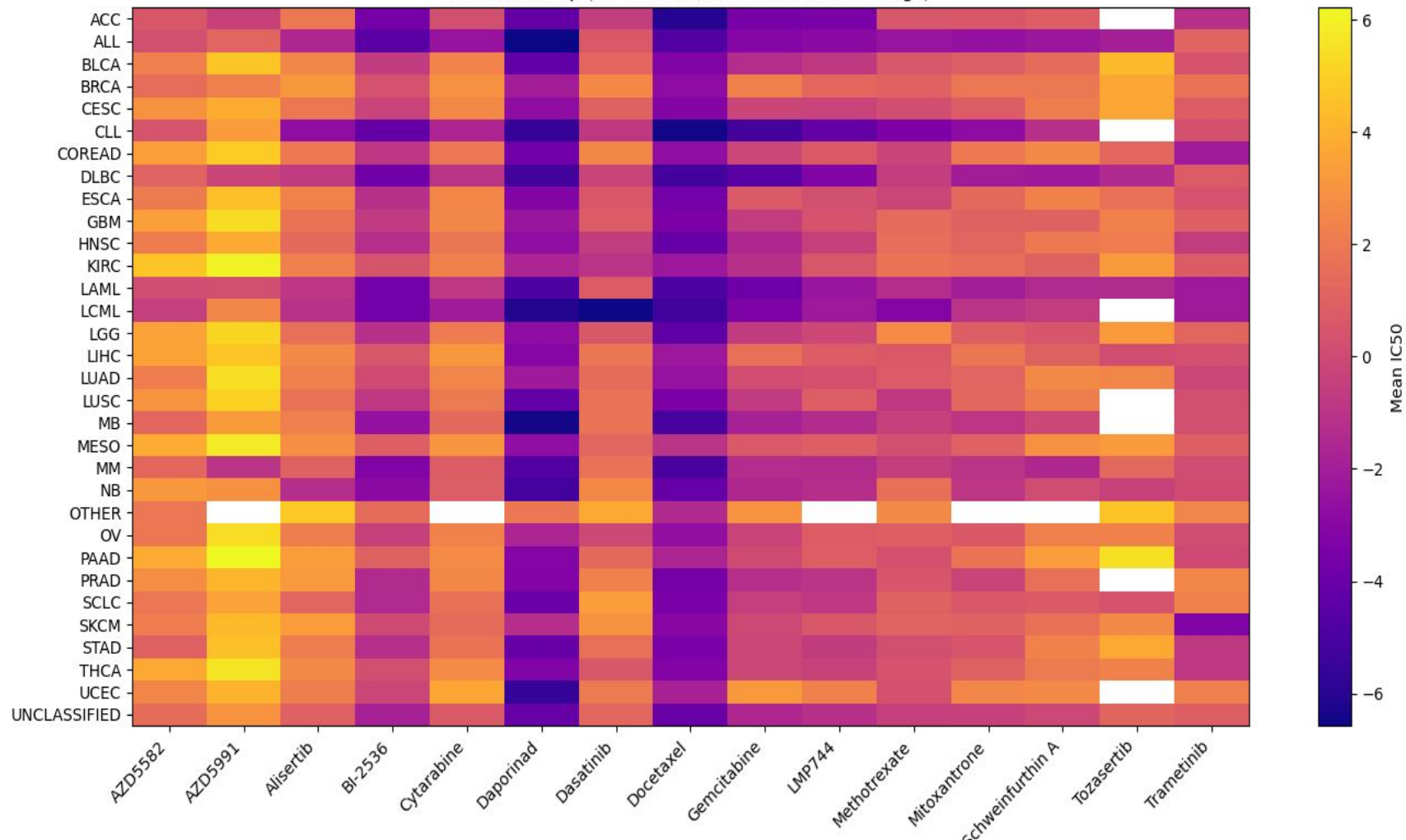
EDA



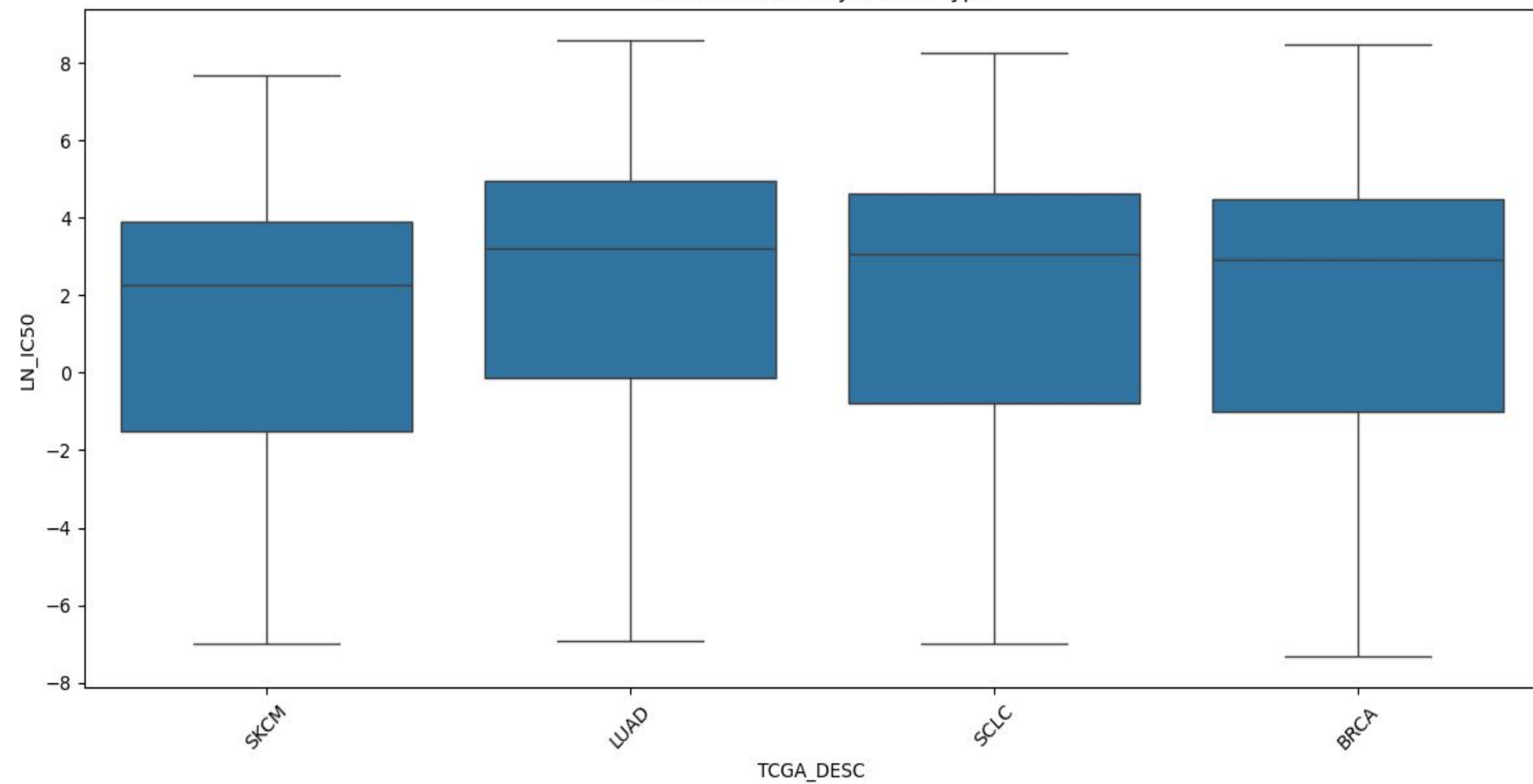
EDA

- ❖ Comparisons of LN_IC50 to different TCGA values and the DRUG_NAME.
- ❖ The lower the LN_IC50 the more sensitive the drug.

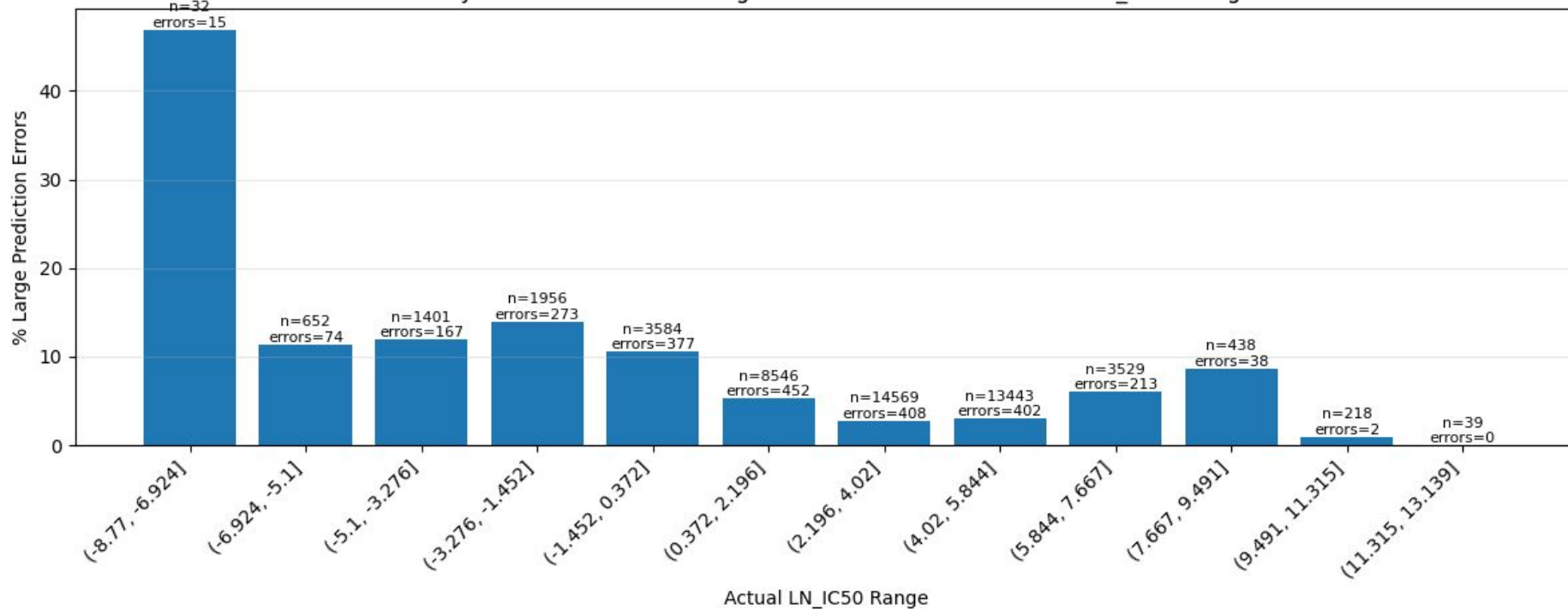
IC50 Heatmap (All Cancers, Most Informative Drugs)



IC50 Distribution by Cancer Type



Final Hybrid Stacked Model: Large Prediction Error Rate Across LN_IC50 Ranges



Preprocessing



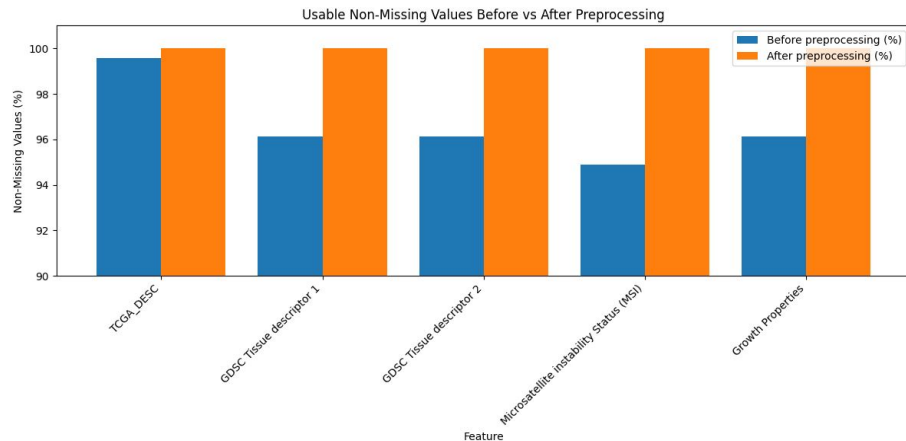
Preprocessing Pipeline Overview

- Main dataset: GDSC drug-response (LN_IC50 as target)
- Supplementary sources: genomic_features.csv, model_list_latest.csv, Model.csv

Key preprocessing phases: Fill missing metadata » Clean TCGA labels »
Add genomic features » Encode & Scale » Train/Test Split

Filling Missing data with supplementary datasets

- MSI and Growth Properties: filled directly from model_list_latest.csv using COSMIC_ID
- GDSC Tissue descriptor 1 & 2: built a lookup table from genomic_features (first non-null per cell line) and used it to fill blanks
- After filling: missingness in tissue descriptors reduced (recover almost all of the missing values)



TCGA Label Cleaning

	A	B	C	D	E	F	G	H	I	J	K
1	COSMIC_ID	CELL_LINE	TCGA_DESC	DRUG_ID	DRUG_NAME	LN_IC50	AUC	Z_SCORE	GDSC Tissue desc	GDSC Tissue desc	Cancer Type
2	683667	PFSK-1	MB	1003	Camptoth	-1.46389	0.93022	0.433123	nervous_system	medulloblastoma	MB
3	684057	ES5	UNCLASSII	1003	Camptoth	-3.36059	0.791072	-0.59957	bone	ewings sarcoma	
4	684059	ES7	UNCLASSII	1003	Camptoth	-3.36059	0.791072	-0.59957	bone	ewings sarcoma	
5	684062	EW-11	UNCLASSII	1003	Camptoth	-3.36059	0.791072	-0.59957	bone	ewings sarcoma	
6	684072	SK-ES-1	UNCLASSII	1003	Camptoth	-5.14296	0.582439	-1.57002	bone	ewings sarcoma	
7	687448	COLO-829	SKCM	1003	Camptoth	-1.23503	0.867348	0.557727	skin	melanoma	SKCM
8	687452	5637	BLCA	1003	Camptoth	-2.63263	0.834067	-0.20322		bladder	
9	687455	RT4	BLCA	1003	Camptoth	-2.96319	0.821438	-0.3832	urogenital_syster	Bladder	BLCA

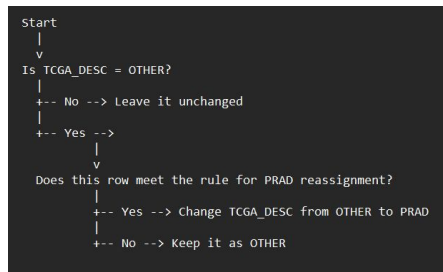
Tissue 2 preprocessed df1

Stage 1 – Fill raw blanks:

- Used from Cancer Type (matching TCGA label) column
- Manual cell-line fixes (e.g., MHH-CALL-4 ➔ ALL)
- Unique 1-to-1 mappings from GDSC Tissue descriptor 2

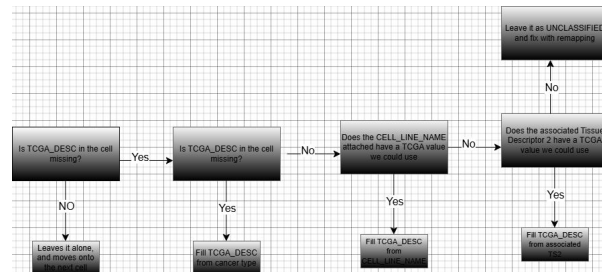
Stage 2 – Resolve ambiguity:

- OTHER ➔ PRAD



Stage 3 – Resolve UNCLASSIFIED:

- Build mapping dictionaries from known-label cell lines (Oncotree lineage, primary disease, subtype, cancer_type)



Adding & Filtering Genomic Features

- Pulled genomic features from model_list_latest: mutational_burden, ploidy_snp6, ploidy_wes
- Kept only columns with $\geq 20\%$ coverage $\Rightarrow$ final set: mutational_burden, ploidy_snp6, ploidy_wes
- Merged on COSMIC_ID; missingness in these columns now minimal ($< 1\%$)
- Remaining missing numeric values imputed with median (full dataset, before split)

Encoding & Scaling Strategy

- K-Fold cross-validated target encoding (5-fold) with smoothing applied to DRUG_NAME to prevent overfitting
- Remaining categoricals (TCGA_DESC, TARGET_PATHWAY, tissue descriptors, MSI, etc.) » one-hot encoding (missing values already filled with explicit “MISSING” category)
- Numeric columns: mutational_burden, ploidies, target-encoded drug » QuantileTransformer (output = normal) fitted on train, applied to test
- All transformations done after train/test split to prevent leakage

Feature Selection



Motivation for Feature Selection

- Dataset expanded to 146 features after encoding
- This high dimensionality increases:
 - Training time
 - Model complexity
 - Risk of overfitting
- Many features contribute minimal predictive signal
- Goals :
 - Identify and retain only the most informative predictors for LN_IC50
 - Remove irrelevant or noisy variables
- This will:
 - Improve model performance and generalization
 - Reduce overfitting and computational cost

Feature Selection Methods

- Baseline approach: Statistical selection (F-test / SelectKBest)
- Final Model approach: Model based selection (XGBoost importance)
- Selection performed after preprocessing and encoding
- Features ranked based on:
 - Relationship with LN_IC50
 - Contribution to prediction accuracy
- Baseline result : top 20 features
- Final Model result : ~ 70 features

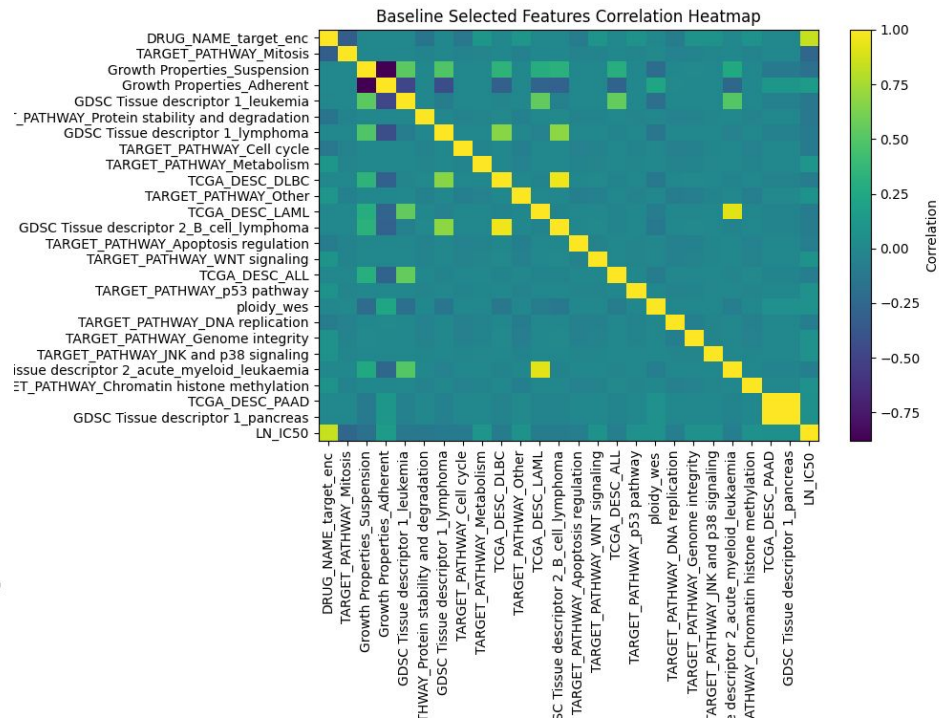
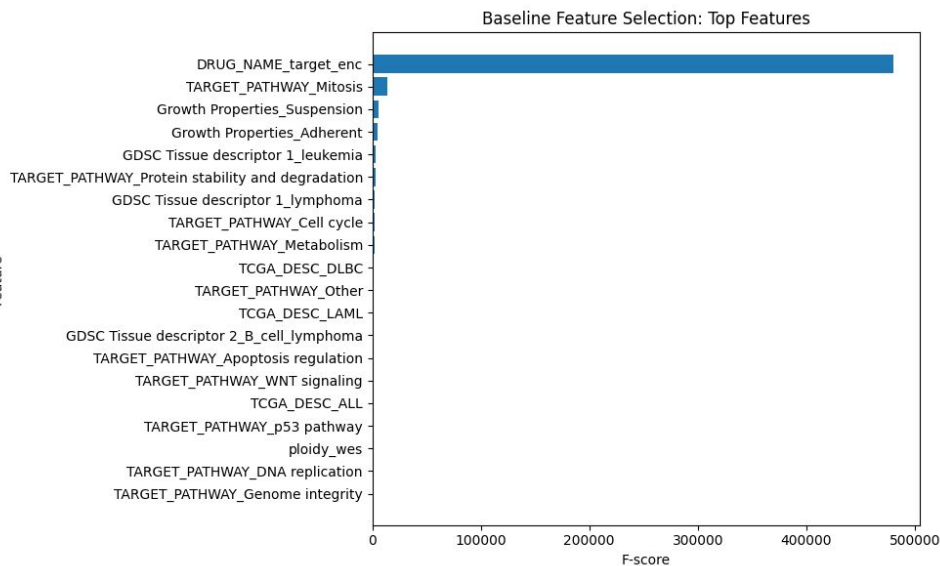
Important Features

- Drug related features had highest importance:
 - DRUG_NAME_target_enc (strongest predictor)
 - TARGET_PATHWAY
- Genomic features contributed meaningful signal:
 - mutational_burden
 - ploidy_wes, ploidy_snp6
- Cancer context features added supporting information:
 - TCGA_DESC
 - Tissue descriptors

Impact of Feature Selection

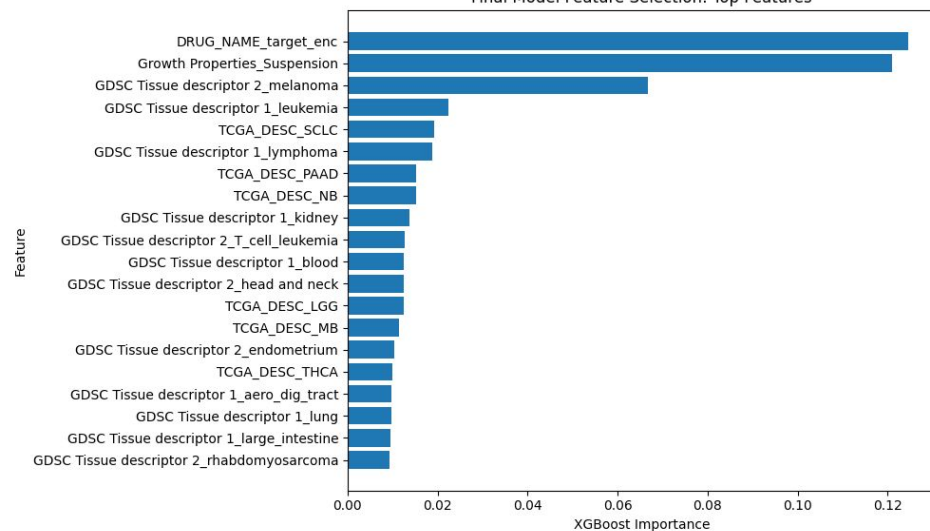
- Reduced feature space —> faster training time
- Improved model stability and generalization
- Helped gradient boosting models perform better
- Reduced noise from weak or redundant features
- Maintained strong predictive accuracy ($R^2 \sim 0.86$ with tuned models)

Baseline Feature Selection Plots

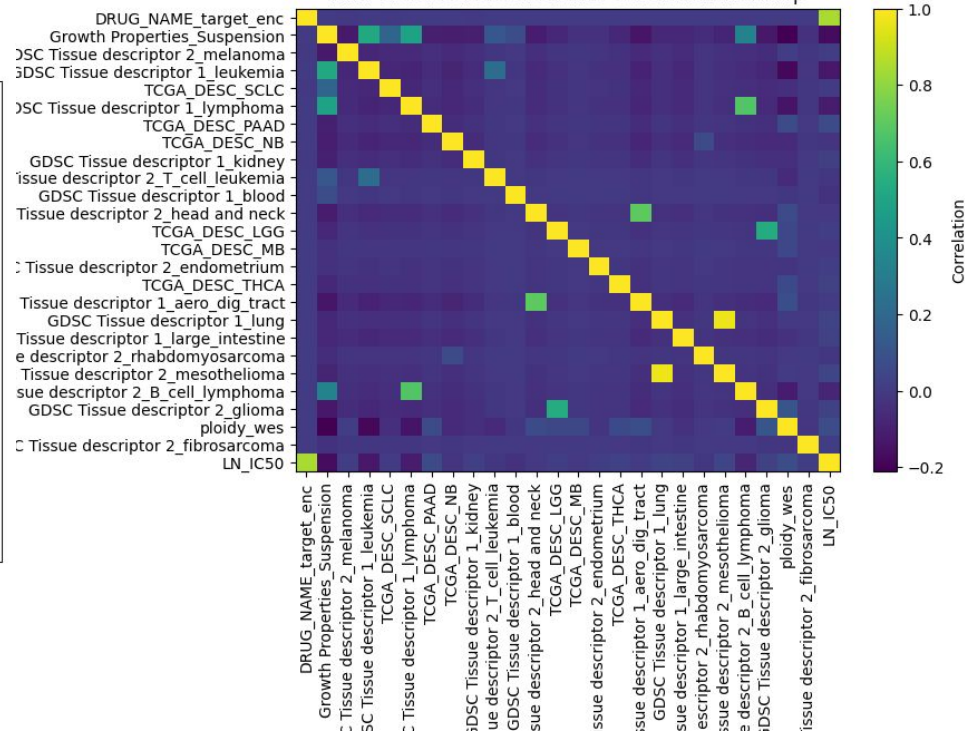


Final Model Feature Selection Plots

Final Model Feature Selection: Top Features



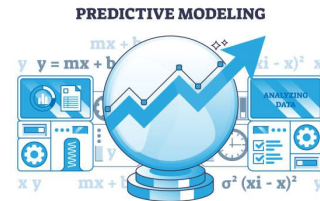
Final Model Selected Features Correlation Heatmap



Models



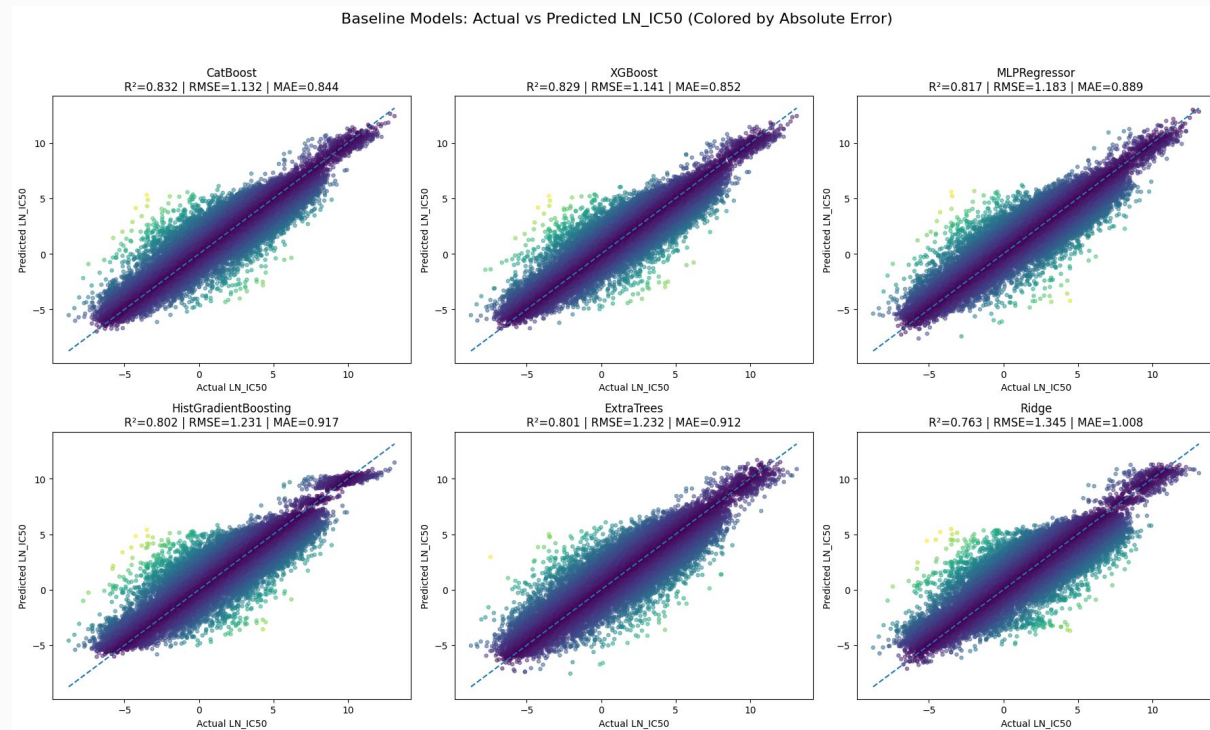
Baseline



- The Baseline tested several regression models to see whether LN_IC50 score could be predicted from the cleaned preprocessed data. This step helped us to have starting performance that we could further advance.
- The performance table below shows that boosting models performed the best, with CATBOOST and XGBOOST leading the baseline results. While other models somewhere near the top scores, RIDGE performed the worst under baseline conditions.

Model	RMSE	MAE	R ²
CatBoost	1.132169	0.843563	0.831986
XGBoost	1.140523	0.851656	0.829497
MLPRegressor	1.182636	0.888743	0.816673
LightGBM	1.229644	0.916978	0.80181
HistGradientBoosting	1.230592	0.916503	0.801504
ExtraTrees	1.232253	0.911635	0.800968
Ridge	1.344522	1.007612	0.763048

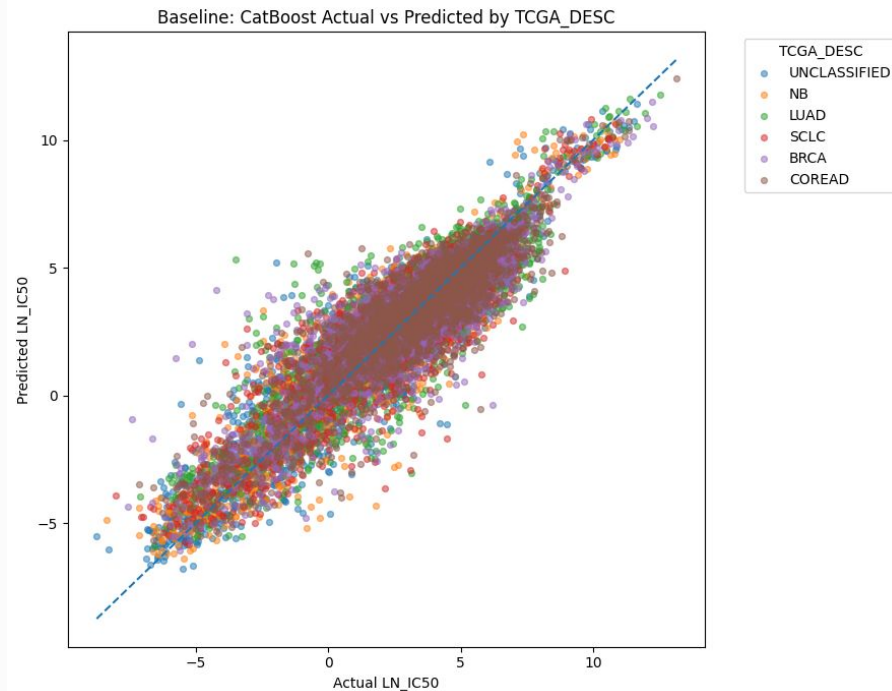
Baseline Model Results



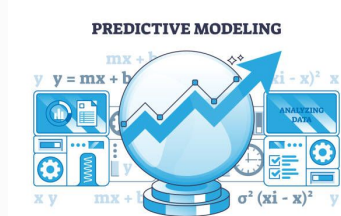
- The figure shows that CatBoost and XGBoost performed strongest in the baseline stage, while Ridge was weaker by checking the prediction point correlation with the actual.
- The top-right part of each plot shows the most differences among the 6 models displayed.

Baseline Model Correlations

- The actual vs Predicted plot shows that a huge portion of the COREAD or Colon Adenocarcinoma, is the most heavily represented TCGA group.
- The BRCA or Breast Invasive Carcinoma, shows more points that fall farther from the diagonal, suggesting less precise prediction for some breast-cancer samples.



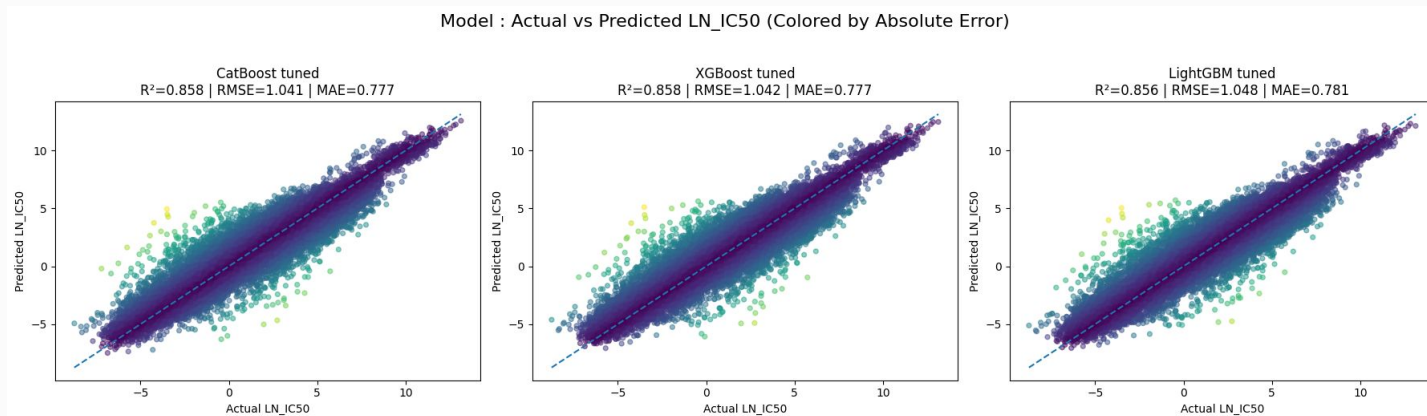
3-Model Approach



- The 3-model approach focused on the three strongest boosting models from the earlier testing: CatBoost, XGBoost, and LightGBM. This step helped us compare the top performing models side by side after using Optuna tuning to improve their parameters.
- The performance table below shows that all three tuned models performed very similarly, with CatBoost having the best score by a small margin. This suggests that the cleaned feature set worked well across multiple boosting models, not just one specific algorithm.

Model	RMSE	MAE	R2
CatBoost tuned	1.041411	0.777037	0.857843
XGBoost tuned	1.041933	0.777403	0.8577
LightGBM tuned	1.047572	0.781412	0.856156

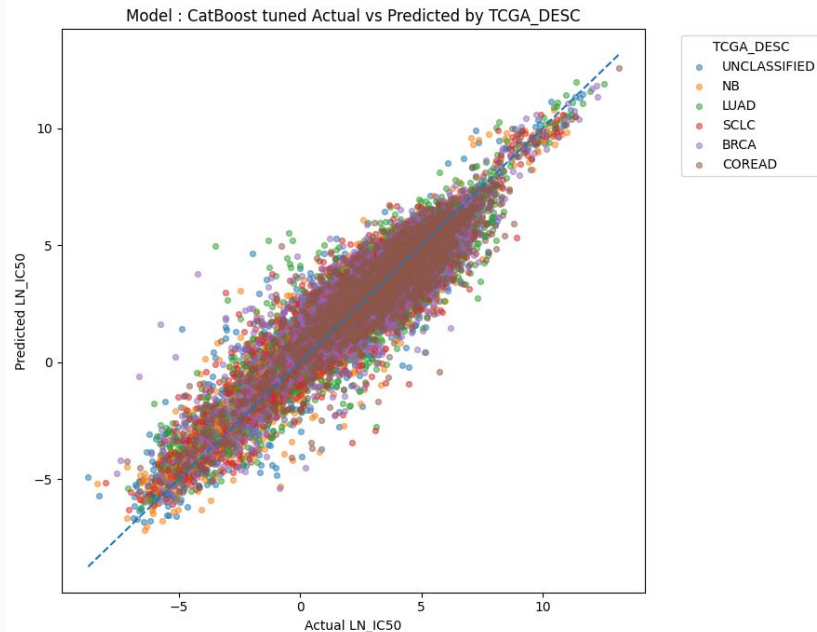
3-Model Results



- The figure shows that all 3 tuned models performed very similarly, with CatBoost and XGBoost slightly stronger based on their closer prediction trend and slightly better R^2 , RMSE, and MAE values.
- The main differences appear in the wider scattered points away from the diagonal line, especially around the middle LN_IC50 range, where prediction errors are more visible.

3-Model Correlations (with CatBoost)

- The actual vs. predicted plot shows that COREAD, or colon adenocarcinoma, is the most heavily represented TCGA group, with many points clustered near the diagonal. BRCA, or breast invasive carcinoma, also stays relatively close to the diagonal, suggesting stronger prediction consistency for many breast-cancer samples.
- LUAD and UNCLASSIFIED have more points farther from the diagonal this time, suggesting less precise prediction for some lung adenocarcinoma and unclassified samples for this model.

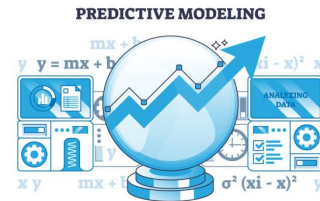


Since the 3-model approach gave a better score, should we leave it as is?

The 3-model approach gave a strong score, so it is a good benchmark, but leaving it there would only show which individual model performed best. Doing more with it is useful because the models may capture different parts of the cancer-drug response pattern, so combining or comparing their outputs can test whether the final prediction becomes more stable, reliable, and less dependent on one model's weaknesses.



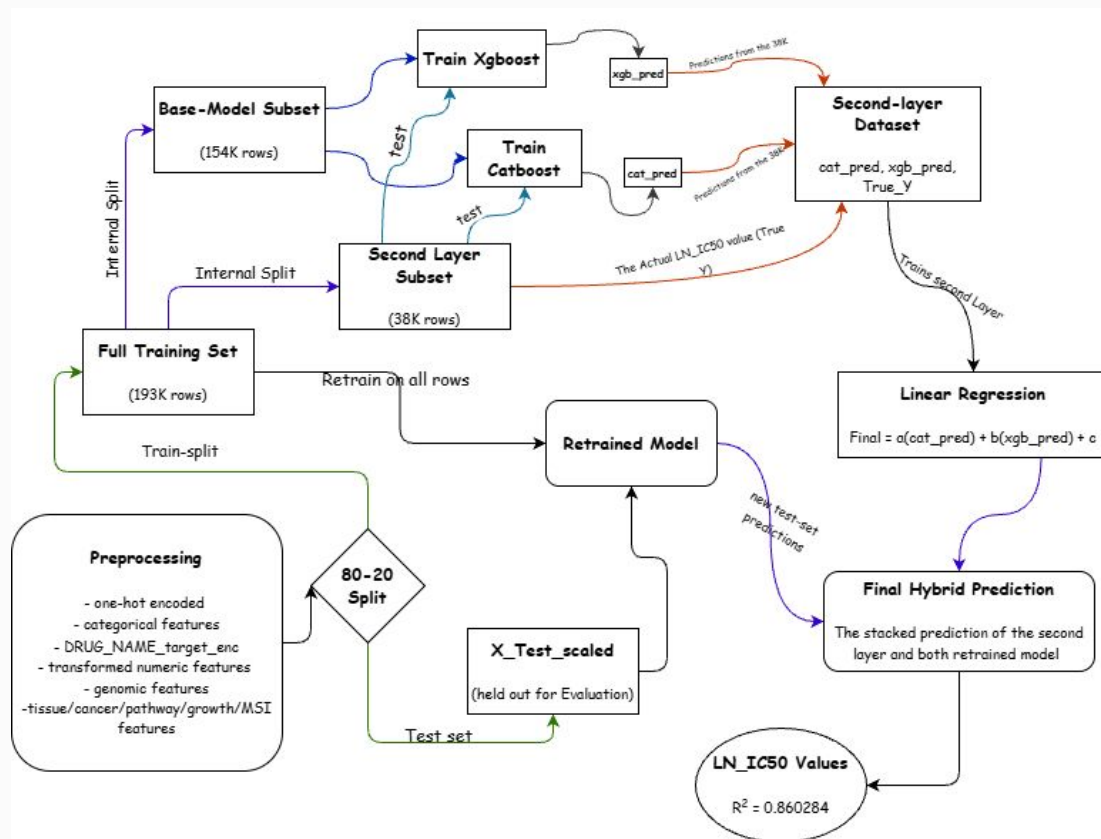
Hybrid Approach



- The Hybrid approach built on the strongest models by combining CatBoost and XGBoost predictions instead of choosing only one final model. This step helped us test whether using two strong model outputs together could create a more stable final prediction.
- The performance table below shows that the stacked model performed the best overall, with the highest R^2 and lowest error. Even though the improvement was small, it showed that combining model predictions can slightly improve performance and reduce reliance on one model alone.

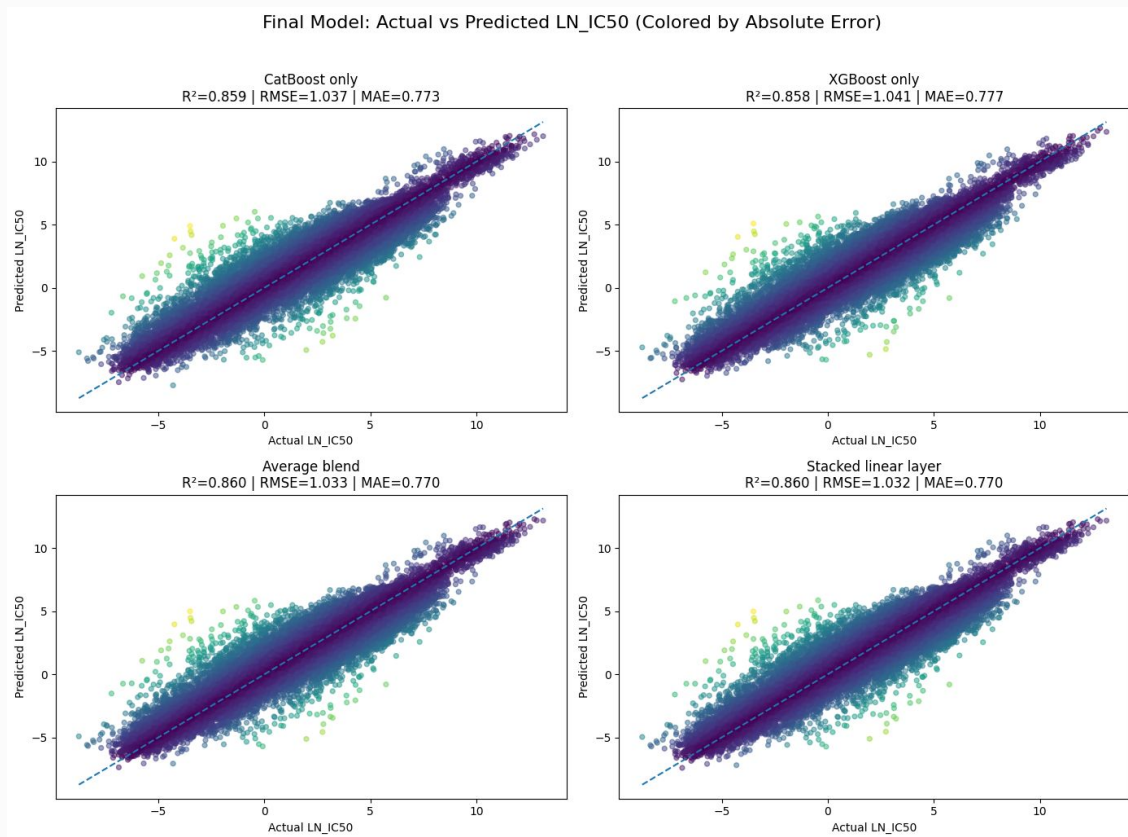
Model	RMSE	MAE	R^2
Stacked linear layer	1.032433	0.769603	0.860284
Average blend	1.032626	0.769966	0.860231
CatBoost only	1.036871	0.772769	0.85908
XGBoost only	1.041301	0.77674	0.857873

Hybrid Approach



- The cleaned dataset was split into an 80/20 train-test split.
- The training set was internally split into a base-model subset and a second-layer subset.
- CatBoost and XGBoost were trained on the base-model subset.
- Both models predicted LN_IC50 on the second-layer subset.
- Those predictions, plus the true LN_IC50 values, were used to train a linear regression second layer.
- The linear regression formula combines both predictions as: **final prediction = a(CatBoost prediction) + b(XGBoost prediction) + c**.
- In the formula, a and b are the learned weights for CatBoost and XGBoost predictions, while c is the intercept that adjusts the final LN_IC50 prediction.
- CatBoost and XGBoost were then retrained on the full training set.
- The retrained models predicted the held-out test set.
- The second-layer linear model combined both predictions into the final hybrid LN_IC50 prediction.

Hybrid Model Results



- The figure shows that the stacked linear layer and average blend performed slightly better than CatBoost and XGBoost alone, based on their tighter prediction trend and slightly higher R^2 with lower RMSE and MAE.
- The main differences appear in the middle LN_IC50 range, where the stacked and blended models show slightly fewer large error points away from the diagonal line.

Hybrid Model Correlations



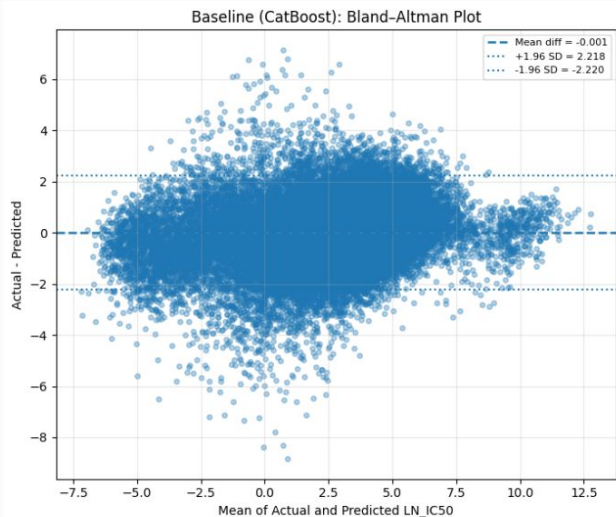
- The final stacked model plot looks very similar to the 3-model TCGA plot, with COREAD and BRCA staying heavily clustered near the diagonal.
- LUAD and UNCLASSIFIED still show more points farther from the diagonal, meaning those groups remain harder to predict precisely. But for both 3-model and the hybrid, it is more closer to the diagonal line.
- Compared with the baseline to 3-model jump, the difference between the 3-model and final hybrid plot is less visually obvious, suggesting the final model gave a smaller performance refinement rather than a major shift in TCGA/LN_IC50 prediction patterns.

How is this Hybrid approach helpful and why chose this above the 3-model approach?

The hybrid model is better because it combines CatBoost and XGBoost predictions through a second layer instead of relying on one model alone. This makes it more stable than the baseline or 3-model approach and better captures complex cancer-drug response patterns.

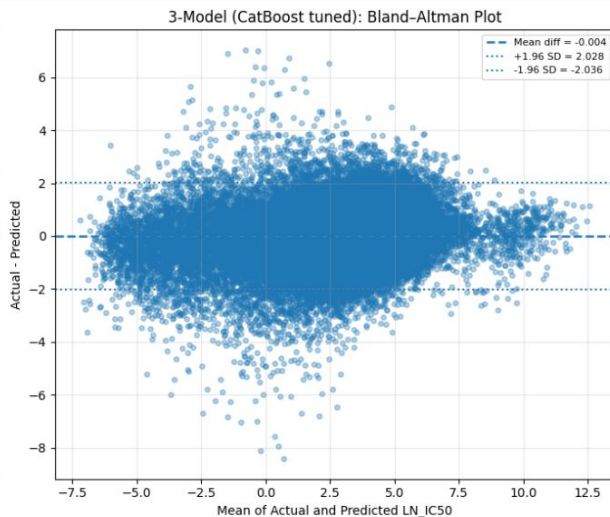


Differences between the Models (bland-altman plots)



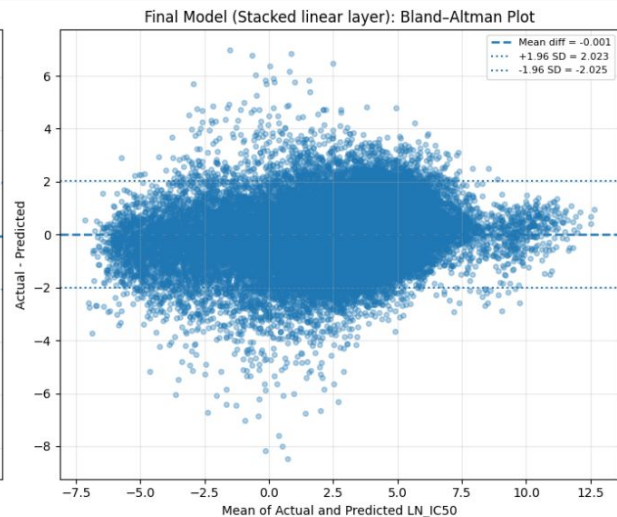
BASELINE

Errors are centered near zero, but the spread is widest, so predictions are less consistent.



3-Model Approach

Errors stay centered near zero and are more tightly grouped than baseline.



Final Model (Hybrid Approach)

Errors remain centered near zero with the tightest spread, showing the most consistent predictions.

Final Output



Dashboard

- The final aspect of our project is to design a dashboard that is intended to be interactive and a tool to assist healthcare professionals in analyzing cancer drug response. The goal is to allow users to input a cancer related profile and receive both predicted drug responses and a ranked comparison of different treatment options.
- The dashboard also includes model comparison, prediction-strength feedback, feature-influence summaries, what-if analysis for selected biological inputs, and chart-based views of ranked drug response patterns to make the predictions more interpretable and easier to compare in context.

When poll is active respond at Pollev.com/emnoon



Through data science methods and cancer pharmacogenomics we were able to show the relationship between drug sensitivity, genomic characteristics and tissue context to provide personalized cancer treatment plans to increase the efficacy of treatment.

Any Questions?

Sources

DIVIDING SECTIONS

Intro **JULIE** - basic bio background relevant to this project, objectives of our project (don't use acronyms without explaining what they mean), problem statement

Data Science aspect/relevant work **GRACE** (why even do the project) - (also, answer questions with graphs/plots) explain where current work exists in domain of research- explain need/ how our project is different than others

Data Acquisition **EM, JULIE** (how we got everything) - source of data (sanger) and explanation of how that data was collected, what they were measuring (explanation of variables) how are we using this data to attain the objectives of our project?

EDA **GRACE, EVERYONE** - findings from the exploration of the nature and attributes of the data- add plots- distributions, correlations, unique values, etc. categorical vs numerical attributes

Preprocessing **HIU, MOE** - techniques used to process the data and how we changed the original data- what we added, what we took away, and why (innovation) normalization(standard scalar to quantile), encoding

Feature Selection **EM, MOE** - how was feature selection done (techniques used, machine learning

Model **ALLAN, HIU, EVERYONE** - run through the models we tested (innovation), metrics of accuracy, what they mean, solutions (hybrid) explain any machine learning, result explanation

Present baseline in one slide, present next few models in next 2 or 3 slides. Give feedback on why specific models may perform better. Then another slide with final, 2-layer model. In separate slide, present evaluation metrics, and in another slide, results. (present results all together)

Discuss hyperparameter choices as well

Final Output **ALLAN** (DASHBOARD) - tour of the final dashboard- what it does,

Outro **JULIE** - 1 slide for how it answers the questions/achieves the objectives that we set out at the beginning- and this is what we have learned- and asks, "any questions?"

****Innovation - means any challenges we faced and how we dealt with them

The final project demo is intended to showcase the end result of the team's effort towards the project over the period of the semester a

The time between the three project demos in this course is an opportunity for teams to resolve any impending issues in design/implementation, add extra features, and/or improve presentation of the insights gained into the problem as a compelling data story.

Requirements:

Project:

The team should have addressed any concerns raised or suggestions/feedback provided by the instructor on previous project demos and the interactive dashboard design.

Presentation:

The presentation should be ~15-20 minutes long leaving ~5-10 min for Q&A. Assume that your expected audience is well-aware of the fundamentals of Data Science, but does not necessarily know about advanced machine learning techniques.

The following tips may be considered while working to create the presentation:

Think carefully about how you would divide the ~20 minutes into two (potentially interspersed) parts: (a) presentation of concepts/ideas and (b) the demo of the working model (maybe add a dashboard or infographics).

As a rough guideline: A 50-50 time split for the presentation and demo sounds like a good split, but you will decide what best suits your work.

Aim for a reasonable number of slides based on the time you have set aside for the concepts/ideas presentation: one sparse-text slide per minute is a good rule of thumb.

It may be easier to start the outline of the presentation with
objectives of the project
findings from exploration of the nature and attributes of data
techniques to process data
machine learning technique(s) applied
performance with appropriate metrics

which correspond to the contents of the project charter, data report, model report, and performance report that you have already created.

If you find that some slides/content would add value to the presentation but are pressed for space - e.g. answers to natural questions the audience might have, alternative methods tried, etc. - move the content to a section called 'Expected Questions' at the end of the slide deck. This way, you can save time by not presenting all the information upfront, but are also prepared in case anyone is interested to know.

Use this template

